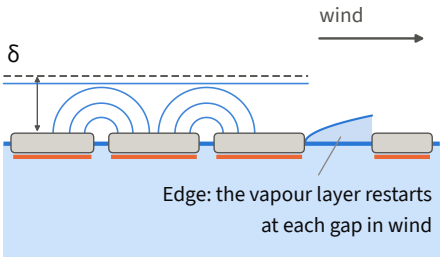
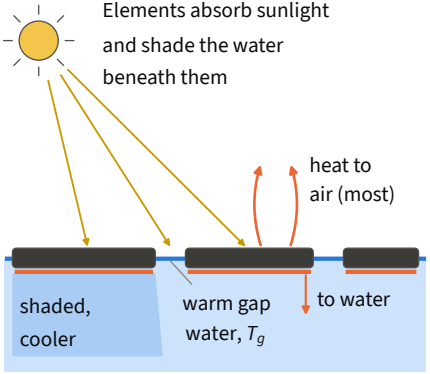
1 Gap evaporation

Gaps narrower than δ: vapour shells merge below δ, so the flux per open area rises



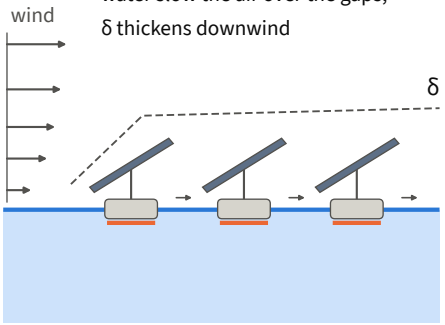
2 Energy and thermal feedback

Elements absorb sunlight and shade the water beneath them



3 Aerodynamic sheltering

Elements standing above the water slow the air over the gaps; δ thickens downwind



$$\frac{S}{\phi} - 1 = \frac{1-\phi}{\phi}(1 - r_{\text{open}}) - r_{\text{under}}$$

open water — covered area

open water, rate $r_{\text{open}}\,E_o$

under the cover, $r_{\text{under}}\,E_o$

Term moved and usual sign

$r_{\text{open}} > 1$ so $S/\phi < 1$	$r_{\text{open}} < 1$ shading cools the gaps	$r_{\text{open}} < 1$ S/ϕ can exceed 1
The excess grows as r/δ falls below about 3; none for gaps much wider than δ	$r_{\text{open}} > 1$ heat from elements, insulation, storage	Forced convection only: no effect in calm air
Edges add to it in wind	$r_{\text{under}} > 0$ heat and vapour through porous or translucent covers	Under suspended covers, also lowers r_{under}